Guglielmo Reggio

PhD Candidate, Computational Neuroscience

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Research Interests

Computational models of how the brain forms cognitive maps from first-person experience: egocentric–allocentric reference frame transformations, spatial memory and imagery, and the relation between navigation in physical space and in abstract, conceptual spaces.

Education

2023–2027 **PhD in Cognitive Neuroscience**, *UCL Institute of Cognitive Neuroscience*, London, UK, *expected 2027*
Grid cells and hexadirectional modulation; computational models of spatial cognition. Supervisor: Prof. Neil Burgess. Upgraded from MPhil to PhD in April 2025.

2020–2021 **MSc Machine Learning**, *University College London*, London, UK, *Merit*
Thesis: "Manifold Alignment in Hyperbolic Space". Supervisor: Prof. Bradley Love.

2017–2020 **BSc Physics with Theoretical Physics**, *King's College London*, London, UK, *First-Class Honours*
Thesis: "Cellular Computing & Artificial Cells". Supervisor: Prof. Mark Green.

2014–2017 **BA Philosophy**, *Università Vita-Salute San Raffaele*, Milan, Italy, *108/110*
Thesis: "When Matter Thought Itself". Supervisor: Prof. Gianvito Martino.

Publications

2026 **Reggio G.***, Kim M.*, Ji Z., Bush D., Barry C., Burgess N. Firing rate adaptation in grid cells produces macroscopic hexadirectional modulation. *bioRxiv* (2026). doi:10.64898/2026.10.19.760191

Conference Contributions

2025 **Reggio G.***, Kim M.*, Ji Z., Bush D., Barry C., Burgess N. Hexadirectional modulation of grid cell firing by firing rate adaptation. Extended abstract and poster, *Cognitive Computational Neuroscience (CCN)*, Amsterdam, Netherlands, 12–15 August 2025.

2024 **Reggio G.***, Kim M.*, Ji Z., Bush D., Wang C., Barry C., Burgess N. Hexadirectional modulation of grid cell firing in rodents. Poster, *iNAV Interdisciplinary Navigation Symposium*, Merano, Italy, 17–21 June 2024.

** Equal contribution.*

Meetings attended

2026 7th International Conference on the Mathematics of Neuroscience and AI, Rome, Italy

2023 Neuro-AI Talks (NEAT), Osnabrück, Germany

Research Experience

2023–present **Doctoral Researcher, Space & Memory Group**, *UCL Institute of Cognitive Neuroscience*, London, UK
Showed that firing rate adaptation provides a plausible mechanism for hexadirectional modulation, combining recordings of almost 4,000 grid cells with simulations of grid cell populations. Interested in egocentric–allocentric transformations and cognitive maps.

2022–2023 **Research Assistant**, *UCL Queen Square Institute of Neurology, with the Space & Memory Group*, London, UK
Firing rate adaptation and hexadirectional modulation in rodent grid cell recordings. Supervisor: Prof. Neil Burgess.

Summer 2020 **Research Intern**, *Burak Lab, The Hebrew University of Jerusalem*, Remote
Ring attractor network model of place cells. Supervisor: Prof. Yoram Burak.

Summer 2019 **Summer Material Analyst**, *Fermi National Accelerator Laboratory*, Batavia, IL, USA
Surface roughness in nitrogen-doped SRF cavities. Supervisor: Dr Anna Grassellino.

Summer 2019 **Research Fellow**, *Randall Centre for Cell & Molecular Biophysics, King's College London*, London, UK
Turing patterns in reaction-diffusion systems on complex networks. Supervisor: Dr Attila Csikász-Nagy.

Teaching and Supervision

2026–present **Literature Project Supervisor**, *UCL Institute of Cognitive Neuroscience*, London, UK
Third-year MSci Neuroscience literature project on vector codes beyond physical space, with Prof. Neil Burgess.

2023–present **Postgraduate Teaching Assistant**, *UCL*, London, UK
NEUR0016 Neural Computation: Models of Brain Function (final-year undergraduate and MSc). Tutorials and assessment marking.

2023–present **Executive Training Demonstrator**, *Division of Psychology & Language Sciences, UCL*, London, UK
Executive education in behavioural science (UCL Changing Minds, UCL Behavioural Insights Hub).

Awards & Funding

2023–2027 Executive Training Demonstrator Studentship, UCL Division of Psychology & Language Sciences (PhD fees and stipend)

2024 Guarantors of Brain Travel Award (iNAV 2024)

2019 King's Experience Research Award, King's College London

Other Experience

2025–present **Co-Founder & Director**, *The Collective Ventures Ltd*, London, UK
Music and arts events.

2019–2020 **Co-Founder & CEO**, *Mappin Technologies Ltd*, London, UK
Indoor navigation for large venues (Kings20 and Geovation accelerators).

Skills

Programming Python: computational modelling and simulation of neural circuits, statistical modelling of neural data, behavioural trajectory analysis. Also Julia, MATLAB, Fortran; Git, LaTeX, Linux.

Languages Italian (native), English (fluent), French (working proficiency)